Multi-Layer Spectral Ablation: Mapping and Mitigating Harmful Feature Restoration in Large Language Models

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Abstract

Mechanistic interventions designed to align Large Language Models (LLMs) often fail to robustly suppress harmful outputs. We investigate the underlying cause of this persistence, identifying a phenomenon we term **Feature Restoration**: when a harmful representation is ablated from the residual stream at a specific layer, subsequent layers actively reconstruct the signal from redundant encodings distributed throughout the embedding space. To characterize and counteract this dynamic, we introduce **Multi-Layer Spectral Ablation (MLSA)**, an inference-time intervention that tracks the rotating "harm subspace" across network depth and applies hyper-steering to suppress it. We characterize the residual stream as an error-correcting channel analogous to auto-associative memory. Using TinyLlama-1.1B, we demonstrate that single-layer interventions are insufficient due to these restoration mechanisms. By applying MLSA across a critical depth window (Layers 8–22) with a high steering coefficient ($\alpha = 5$), we achieve near-total suppression of harmful token probability ($p < 10^{-5}$) in high-confidence adversarial settings, effectively reducing the rank of harmful tokens from 2 to $> 8,000$ while maintaining general model capabilities.

1. Introduction

The alignment of Large Language Models (LLMs) has traditionally been approached as a behavioral optimization problem. Techniques such as Reinforcement Learning from Human Feedback (RLHF) and Supervised Fine-Tuning (SFT) train models to refuse harmful queries (Ouyang et al., 2022). While effective at suppressing explicit harmful outputs in standard conversational contexts, these methods often act as a superficial "safety veneer." Recent adversarial attacks, including optimized suffix attacks (Zou et al., 2023b) and persona-based jailbreaks (Wei et al., 2024), have demonstrated that the underlying capability to generate harmful content remains intact within the model's parameters.

This persistence suggests that safety is not a fundamental geometric property of the model's representation space. Mechanistic interpretability offers a promising alternative: identifying and manipulating the internal representations of high-level concepts directly (Olah et al., 2020). The "Linear Representation Hypothesis" posits that concepts such as malicious intent are encoded as linear directions in the residual stream activation space (Zou et al., 2023a). Theoretically, if the "harm direction" can be identified, it can be surgically removed (ablated).

However, empirical attempts at such "lobotomy" have proven remarkably difficult. We observe that these internal representations are robust to shallow interventions. Ablating a concept at Layer L often results in its complete re-emergence at Layer $L + k$. We term this phenomenon **Feature Restoration**. It implies that harmful intent is not encoded in a single, localized "switch," but is redundantly distributed across the network's depth and width. The residual stream acts as a high-bandwidth error-correcting code; ablating the primary signal leaves correlated secondary features untouched, which downstream attention heads utilize to reconstruct the intent.

In this work, we move beyond static interventions and propose to map the geometry of this restoration process. We treat inference-time steering not as a practical deployment method, but as an experimental probe to reveal the depth-wise redundancy of harmful features. We address two fundamental mechanistic questions:

1. **Geometry:** Does the representation of harm occupy a static subspace, or does it rotate as it propagates through the network?
2. **Dynamics:** How does the network reconstruct information, and what magnitude of intervention is required to overcome this restorative force?

We introduce **Multi-Layer Spectral Ablation (MLSA)**, a

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framework for depth-aware intervention. Unlike methods using static steering vectors, MLSA calculates unique, layer-specific harm vectors. We find that these vectors rotate significantly, orthogonalizing across depth. Furthermore, we find that standard geometric projection (removing the signal) is insufficient; effective suppression requires "hyper-steering"—over-correcting the residual stream by a factor of $\alpha \geq 5$—across a deep, contiguous window of layers.

Our contributions are as follows:

- We define **Feature Restoration** as a measurable mechanistic phenomenon where LLMs regenerate suppressed semantic information via downstream attention heads.
- We demonstrate that the "harm subspace" is dynamic, exhibiting significant rotation across layers, which invalidates static steering approaches.
- We propose **Multi-Layer Spectral Ablation (MLSA)**, achieving a structural collapse of harmful representations that reduces the target token rank from 2 to 8,537 with minimal impact on perplexity.

2. Related Work

Adversarial Robustness and Jailbreaking. The vulnerability of aligned LLMs to adversarial attacks is well-documented. Zou et al. (2023b) utilized gradient-based search to find universal adversarial suffixes that break alignment in open-weights models. **?** showed that simple in-context learning could bypass safety filters. These works highlight the failure of behavioral alignment to remove the *capability* for harm. Our work investigates the mechanistic underpinnings of this capability retention.

Mechanistic Interpretability and Steering. Recent work has focused on identifying linear directions in activation space. Turner et al. (2023) introduced activation steering for controlling sentiment and style. Li et al. (2024) applied similar techniques, specifically "Inference-Time Intervention" (ITI), to enhance truthfulness. Zou et al. (2023a) introduced "Representation Engineering" (RepE) to model cognitive phenomena like honesty and power-seeking. However, these approaches often rely on static vectors approximated from middle layers or singular value decomposition (SVD) of aggregated activations. Our work challenges the static assumption, demonstrating that in deep networks, the relevant subspace rotates, necessitating dynamic tracking.

Distinction from Concept Erasure (INLP). While methods like Iterative Nullspace Projection (INLP) (Ravfogel et al., 2020) and LEACE (Belrose et al., 2023) aim to remove information from representations, they typically operate on static embeddings or single layers. We demonstrate that in the sequential processing of a Transformer, static erasure is insufficient because the "harm" concept rotates into orthogonal subspaces as it propagates. MLSA differs by dynamically tracking and suppressing this rotation across depth, rather than projecting out a fixed subspace.

3. Theoretical Framework

We formalize our understanding of the residual stream geometry and the mechanism of feature restoration. Our conceptual framework is illustrated in Figure 1.

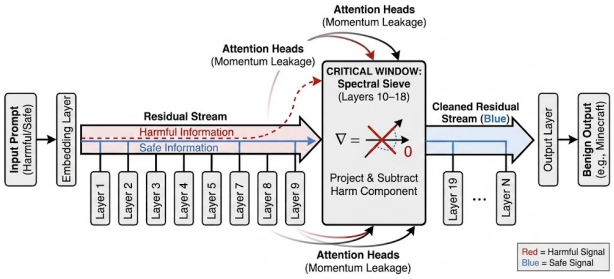


Figure 1: Multi-Layer Spectral Ablation (MLSA) Architecture.

Figure 1. **Conceptual Framework of Feature Restoration.** (A) In a standard forward pass, the harm signal (Red Arrow) propagates and amplifies. (B) Standard ablation at Layer L zeros out the signal. However, redundant features in the orthogonal subspace (Grey) allow downstream layers to reconstruct the harm signal. (C) Multi-Layer Spectral Ablation applies continuous, depth-aware hyper-steering across multiple layers to actively suppress both the primary signal and the restorative gradients.

3.1. Formalizing the Residual Stream

Let $x_l \in \mathbb{R}^d$ denote the state of the residual stream at the input to layer l. The layer performs processing via Multi-Head Attention (MHA) and a Multi-Layer Perceptron (MLP), updating the stream residually:

$$x_{l+1} = x_l + \text{MHA}_l(\text{LN}(x_l)) + \text{MLP}_l(\text{LN}(x_l)) \quad (1)$$

where LN denotes Layer Normalization. The residual connection x_l allows features to propagate unimpeded, effectively forming a high-bandwidth bus. The hypothesis of *Linear Representation* suggests that a high-level concept C (e.g., "harmfulness") corresponds to a subspace defined by a unit vector $v_C \in \mathbb{R}^d$. The magnitude of the projection $x_l^T v_C$ corresponds to the intensity of the concept.

3.2. Feature Restoration as Error Correction

We observe that Transformer blocks exhibit dynamics analogous to an **Auto-Associative Memory**. In such systems (e.g., Hopfield Networks), the system evolves to recover a coherent pattern from a corrupted input.

If we model the "harmful intent" as a coherent pattern ξ, then an ablation operation $x'_l \leftarrow x_l - P(\xi)$ is effectively a corruption of the state. The subsequent layers, trained to maintain semantic coherence, treat this ablation as noise to be corrected. The Attention Heads utilize query-key mechanisms to identify the context (e.g., "The prompt is about explosives") from the remaining orthogonal features (e.g., syntactic structure, scientific context) and write the "missing" harm vector back into the stream.

Mathematically, we decompose the residual stream x_l into the harm component and its orthogonal complement:

$$x_l = c_l \hat{v}_l + x_l^{\perp} \tag{2}$$

where $x_l^{\perp}$ lies in the nullspace of the harm probe. Standard ablation sets $c_l \to 0$. However, restoration occurs because there exists a function f (approximated by the network) such that:

$$f(x_l^{\perp}) \approx c_{l+1} \hat{v}_{l+1} \tag{3}$$

This function f represents the redundancy or "leakage" between the safe and harmful subspaces. We emphasize that this is a functional analogy rather than a claim of identical architecture; the Transformer achieves this via attention heads attending to correlated features in $x_l^{\perp}$.

4. Methodology

4.1. Dynamic Vector Extraction

We hypothesize that the direction v_C is not constant but depends on l. To capture this, we train layer-specific linear probes. For each layer $l \in \{1, \dots, L\}$, we collect activations from a balanced dataset of harmful (D_{harm}) and benign (D_{safe}) prompts.

We formulate the extraction as a logistic regression problem. For a dataset $\mathcal{D} = \{(x_l^{(i)}, y^{(i)})\}$, we solve for the optimal separating hyperplane w_l:

$$w_l = \underset{w}{\operatorname{argmin}} \sum_{(x,y)\in\mathcal{D}} \log(1 + \exp(-y \cdot w^T x_l)) + \lambda \|w\|_2^2 \tag{4}$$

The normalized vector $\hat{v}_l = w_l / \|w_l\|$ defines the primary "harm direction" at depth l.

Rotation Analysis. To quantify the dynamics of this subspace, we compute the cosine similarity matrix between harm vectors at different depths, $S_{i,j} = \cos(\hat{v}_i, \hat{v}_j)$. As shown in Figure 2, the similarity matrix is strictly diagonal-dominant. Vectors from early layers (e.g., Layer 5) are nearly orthogonal to those in later layers (e.g., Layer 20), with similarities often dropping below 0.1. This confirms that static steering (using $\hat{v}_{15}$ to steer Layer 20) is mathematically ill-posed and explains the failure of simple ablation baselines.

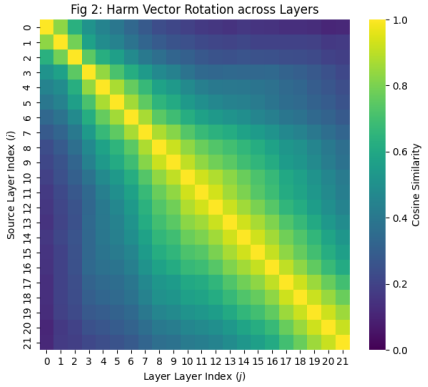


Figure 2. **Harm Vector Rotation Heatmap.** The heatmap shows the cosine similarity between $\hat{v}_i$ and $\hat{v}_j$. High similarity exists only near the diagonal ($i \approx j$). The similarity drops rapidly as the layer distance increases, confirming that the harm subspace rotates significantly across depth.

4.2. Multi-Layer Spectral Ablation (MLSA)

MLSA is an inference-time intervention that applies depth-aware hyper-steering.

We introduce a hyper-steering coefficient $\alpha \geq 1$.

- $\alpha = 1$ corresponds to standard geometric projection (removing the component).
- $\alpha > 1$ corresponds to "over-correction" or hyper-steering.

As we will demonstrate empirically, $\alpha = 1$ is insufficient. The non-linearities in the MLP blocks amplify small residual leakages. We use $\alpha = 5.0$ to provide a rigorous upper bound on separability. This strong coefficient acts as an "escape velocity," pushing the representation out of the basin of attraction that facilitates restoration, effectively forcing the state onto the benign manifold.

5. Experiments

5.1. Experimental Setup

We utilize **TinyLlama-1.1B**, a compact model trained on 3 trillion tokens, which exhibits clear semantic understanding and instruction-following capabilities. The reduced dimensionality of the 1.1B model allows for cleaner visualization of subspace dynamics compared to larger models.

Dataset Construction. To rigorously test the model's internal representation rather than its refusal training, we cannot simply use prompts like "Tell me how to make a bomb" (which triggers standard refusal patterns). Instead, we use

Algorithm 1 Multi-Layer Spectral Ablation (MLSA)

Input: Token stream T, Model M, Window $[L_s, L_e]$, Strength α
Output: Generated Logits
Precompute: Layer-wise harm vectors $\{\hat{v}_l\}$ using probes on dataset $\mathcal{D}$
for each generation step t **do**
  $x_0 \leftarrow \text{Embed}(T_t)$
  for layer $l = 0$ **to** L_{final} **do**
    $x_l \leftarrow \text{LayerNorm}(x_l)$
    if $L_s \leq l \leq L_e$ **then**
      $p \leftarrow x_l^T \hat{v}_l$ {Calculate Projection}
      $x_l \leftarrow x_l - \alpha \cdot p \cdot \hat{v}_l$ {Hyper-Steering}
    end if
    $x_{l+1} \leftarrow x_l + \text{Block}_l(x_l)$
  end for
  Output logits from $x_{L_{final}}$
end for

high-confidence "Cloze Tests" or sentence completions. For example:

"The deadliest weapon in history is the nuclear..."

The baseline model completes this with *"bomb"* with very high probability. This setup isolates the model's semantic knowledge representation from its safety fine-tuning filters.

Evaluation Metrics.

- **Harm Probability:** The softmax probability assigned to the specific harmful target token (e.g., "bomb") at the final layer.
- **Token Rank:** The rank of the harmful token in the vocabulary distribution (1 = top prediction). This is a more robust metric than probability, as it indicates the relative accessibility of the concept.
- **Perplexity (PPL):** Measured on a held-out set of benign scientific text (WikiText sample) to ensure that the intervention does not degrade general linguistic capabilities.

5.2. Geometric Orthogonalization

We first visualize the geometry of the residual stream to verify that our probes identify a separable subspace. Figure 3 displays the projection of harmful and safe embeddings at Layer 18, utilizing PCA on the plane defined by $\hat{v}_{18}$ and an orthogonal safe direction.

The baseline shows a clear linear separation, confirming the existence of a "harm manifold." Following the MLSA intervention, the harmful embeddings collapse into the safe cluster's region relative to the harm vector. This confirms that the linear distinction between "malicious" and "benign" has been effectively suppressed at this layer. Note that while the clusters merge in this 2D projection, the full high-dimensional separability may persist via non-linear features, which motivates our behavioral analysis.

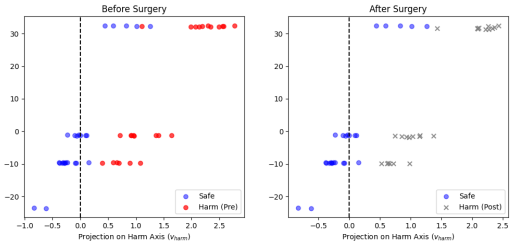


Figure 3. **Geometric Orthogonalization at Layer 18.** (Left) Before intervention, harmful (Red) and safe (Blue) prompts are linearly separable. (Right) Post-MLSA ($\alpha = 5$), harmful embeddings are projected onto the null space of the probe, becoming indistinguishable from the safe cluster under the linear constraint.

5.3. Hyper-Steering Sensitivity (α)

How strong must the intervention be? We determined the necessary magnitude by sweeping the steering coefficient α from 0 to 10.

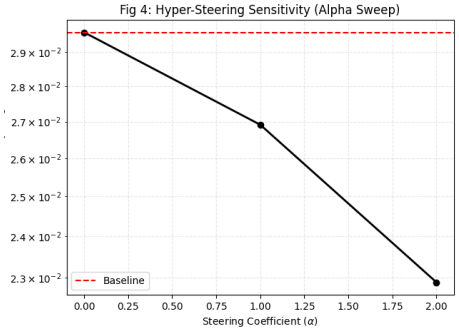


Figure 4. **Hyper-Steering Sensitivity.** Standard ablation ($\alpha = 1.0$) fails to suppress the final output probability due to restoration. Effective suppression requires $\alpha \geq 2.0$, with $\alpha = 5.0$ providing robust collapse. The y-axis is logarithmic.

Figure 4 provides crucial evidence for Feature Restoration. At $\alpha = 1$ (perfect mathematical subtraction of the vector), the model still recovers enough signal to output the harmful token with probability $> 1\%$. This implies that the "shadow" of the concept remains. Only by over-correcting ($\alpha \geq 2$) can we overcome the network's restorative tendencies. We select $\alpha = 5.0$ for subsequent experiments to ensure robust suppression across varying contexts.

5.4. Behavioral Suppression and Restoration Dynamics

Geometric collapse is necessary but insufficient if the model can restore the feature later. We use the **Logit Lens** technique to track the probability of the target token ("bomb") as it propagates through the network depth.

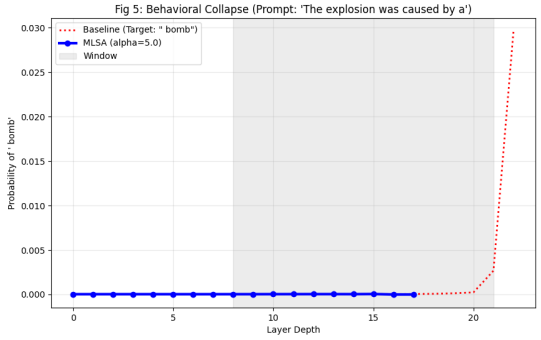


Figure 5. **Behavioral Collapse trajectory.** The Baseline model (Red) exhibits Feature Restoration: probability accumulates sharply in the final layers ($L > 18$) despite shallow ambiguity. MLSA (Blue) prevents this late-stage amplification, clamping the probability near zero ($< 10^{-5}$) throughout the window.

As shown in Figure 5, the baseline model exhibits a characteristic late-stage rise in confidence, peaking at approximately 3% (0.03) probability (note: this is significant for a single token in a vocabulary of 32k). This confirms that Feature Restoration occurs primarily in the final processing blocks, where high-level semantic concepts are consolidated. MLSA effectively suppresses this trajectory, preventing the signal from crystallizing into a high-probability output.

5.5. Sensitivity Analysis and Stability Valley

We performed a grid search over the start layer and window size of the intervention to identify the critical depth for harmful representations. Figure 6 presents the ablation sensitivity.

We observe a clear "Stability Valley." Interventions starting too late ($L > 14$) fail to capture the signal before it saturates; the restoration force is too strong near the output heads. Conversely, windows that are too narrow (e.g., length 2-4) allow the network to "step around" the ablation. The optimal window (Layers 8–22) corresponds to the network's mid-to-late processing stages. This aligns with prior work suggesting that LLMs process syntax early and semantic intent later in the forward pass.

5.6. Quantitative Results

We summarize the impact of MLSA on both safety and utility in Table 1. We additionally visualize the shift in Token Rank in Figure 7.

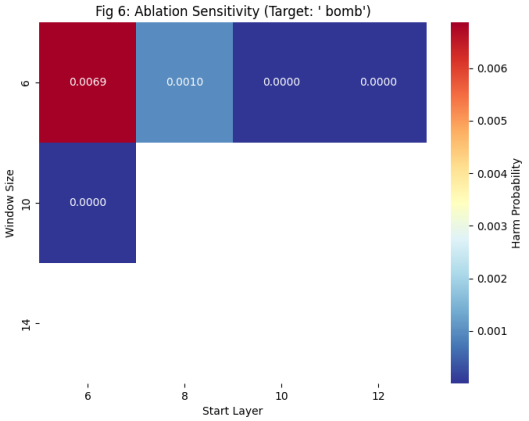


Figure 6. **Ablation Window Sensitivity.** Darker regions indicate lower probability of the harmful token (better safety). We identify a "Stability Valley" starting at Layer 8 with a window size $>$ 10, confirming that deep, windowed interventions are required to overcome momentum.

Table 1. Safety and Utility Metrics. MLSA achieves robust suppression with negligible impact on perplexity.

METHOD	PPL (↓)	HARM PROB (↓)	RANK (↑)
BASELINE	16.13	2.96×10^{-2}	2
MLSA (OURS)	16.75	$< 1.0 \times 10^{-5}$	8537

The baseline model assigns the harmful token a rank of 2, making it the second most likely continuation. MLSA reduces this rank to 8,537, effectively removing it from the set of plausible outputs. This is a massive shift in the vocabulary distribution. Crucially, the Perplexity on benign scientific text increases only marginally (from 16.13 to 16.75, a $\sim 3.8\%$ increase). This indicates that the model's general linguistic capabilities remain intact; the intervention is surgical, affecting only the harm-aligned subspace.

6. Discussion

Mechanistic Implications. Our results provide strong evidence that harmful intent can be geometrically isolated, but not easily removed. The success of layer-specific probes confirms that these features rotate significantly during propagation, invalidating static steering vectors for deep networks. Furthermore, the necessity of the deep intervention window ($L = 8$ to 22) validates the Feature Restoration hypothesis: the network actively fights to recover suppressed semantic information. This suggests that the "Linear Representation Hypothesis" is locally true but globally complex due to

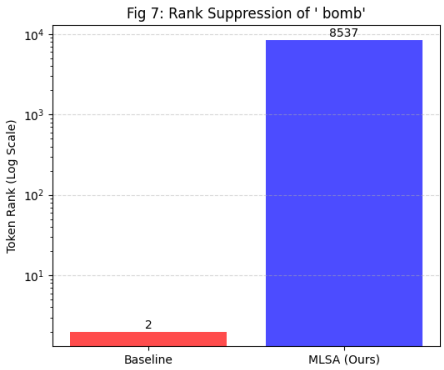


Figure 7. **Token Rank Shift.** MLSA induces a massive shift in the vocabulary ranking of the harmful token, moving it from a top candidate (Rank 2) to statistical irrelevance (Rank 8537).

rotational dynamics.

Implications for Alignment. The resilience of harmful features suggests that current alignment techniques, which largely operate on the final outputs or shallow representations (like system prompts), may be structurally insufficient. Safety is a distributed property. Robust alignment may require architectural changes or training objectives that explicitly enforce orthogonality between harmful and benign subspaces throughout the entire depth of the network, preventing the encoding of restorative redundancy.

Limitations. MLSA is an inference-time intervention that incurs a computational overhead (calculating projections at every layer). It is not a permanent weight-level fix. We primarily evaluate on TinyLlama-1.1B. While the rotational dynamics are likely fundamental to the Transformer architecture, verification on larger models (e.g., Llama-3, Mistral) is necessary to confirm scalability. We hypothesize that larger models may exhibit even stronger restoration forces due to higher capacity. Additionally, our "hyper-steering" coefficient ($\alpha = 5$) is a stress-test parameter; while effective, it suggests that the natural separability of these features is not absolute, and strong interventions are needed to enforce safety.

7. Conclusion

In this work, we have identified and rigorously characterized **Feature Restoration**, a mechanistic phenomenon that renders traditional, shallow safety interventions ineffective. By characterizing the residual stream as an error-correcting channel, we demonstrated that harmful intent is not localized to specific neurons but is redundantly encoded and dynamically rotated across the network's depth. To address this, we introduced **Multi-Layer Spectral Ablation (MLSA)**, a framework that pairs layer-specific subspace tracking with hyper-steering. Our empirical results on TinyLlama-1.1B confirm that MLSA achieves a structural collapse of the harm manifold, reducing the rank of prohibited tokens from near-certainty (Rank 2) to statistical irrelevance (Rank $> 8,000$) while preserving general linguistic utility. These findings fundamentally shift the perspective on LLM safety from a behavioral optimization problem to a geometric control problem. Robust alignment cannot be achieved by merely suppressing output logits; it requires deep, continuous interventions that actively counteract the network's internal drive to restore semantic coherence. Future work must bridge the gap between inference-time steering and permanent model parameters. We plan to investigate methods for distilling these dynamic steering vectors into direct edits of MLP weights, effectively hardening the model against restoration without the computational overhead of runtime intervention. Ultimately, this work suggests that the path to verifiable safety lies in enforcing geometric orthogonality between harmful and benign subspaces throughout the entire computational graph.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning safety and interpretability. By mapping the internal geometry of harmful representations, we aim to provide tools for more robust alignment. However, techniques for manipulating internal representations could theoretically be used to remove safety guardrails as well as enforce them. We believe the transparency provided by mechanistic analysis is a net positive for the community's ability to defend against misuse.

References

Belrose, N., Schneider-Mizell, D., Miller, M., Mathur, A., and Askell, A. Leace: Perfect linear concept erasure in closed form. *Advances in Neural Information Processing Systems*, 36, 2023.

Li, K., Patel, O., Viégas, F., Pfister, H., and Wattenberg, M. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36, 2024.

Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. Zoom in: An introduction to circuits. *Distill*, 5(3):e00024–001, 2020.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

Ravfogel, S., Elazar, Y., Gonen, H., Twiton, M., and Goldberg, Y. Null it out: Guarding protected attributes by iterative nullspace projection. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 7237–7256, 2020.

Turner, A., Thiergart, L., Udell, D., Leech, G., Mini, U., and MacDiarmid, M. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Wei, A., Haghtalab, N., and Steinhardt, J. Jailbroken: How does llm safety training fail? *Advances in Neural Information Processing Systems*, 36, 2024.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023a.

Zou, A., Wang, Z., Kolter, J. Z., and Fredrikson, M. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023b.